

Chapter 11 Study Sheet

Circumference, Area, and Volume — everything you need on one page.

11.1 Circumference and Arc Length

- $C = 2\pi r = \pi d$. Arc length $s = (m/360) \cdot 2\pi r$ — a fraction of the circumference.
- Radians: $180^\circ = \pi$ rad. Multiply degrees by $\pi/180$. In radians, $s = r\theta$.

11.2 Areas of Circles and Sectors

- $A = \pi r^2$. Sector = $(m/360) \cdot \pi r^2$ — the same fraction idea, applied to area.
- Backward problems: plug into the same equation and solve for x , r , or the area.

11.3 Areas of Polygons

- Rhombus/kite: $A = \frac{1}{2}d_1d_2$. Regular polygon: $A = \frac{1}{2}aP$ with the apothem a .
- Apothem right triangle: $\tan(180^\circ/n) = (s/2)/a$. Hexagon shortcut: $a = (s/2)\sqrt{3}$.
- Central angle of a regular n -gon: $360/n$.

11.4 Three-Dimensional Figures

- Euler's Formula: $F + V - E = 2$. Cross sections are flat slices; spheres always slice to circles.
- Solids of revolution: rectangle $\rightarrow$ cylinder, right triangle $\rightarrow$ cone, semicircle $\rightarrow$ sphere.

11.5 Prisms and Cylinders

- $V = Bh$, where B is the base AREA (compute it first!). Cylinder: $B = \pi r^2$.
- Cavalieri's Principle: oblique solids use the same formula — same layers, same volume.

11.6-11.7 Pyramids and Cones

- $V = \frac{1}{3}Bh$ — one third of the matching prism/cylinder. Height h is PERPENDICULAR height.
- Cone: $\ell^2 = r^2 + h^2$; $SA = \pi r^2 + \pi r\ell$ (slant height only appears in surface area).

11.8 Spheres

- $SA = 4\pi r^2$; $V = (4/3)\pi r^3$. Hemisphere: half of each, plus the flat circle πr^2 for total SA.
- Scaling: radius $\times k \rightarrow$ surface $\times k^2$, volume $\times k^3$.
- Composites: split into cylinder + hemisphere (silo) or similar; add the volumes; keep π until the end.